INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

(ICDFA)

Governance, Risk and Compliance (GRC)

Week 1 – Lab 1 Assignment

LINUX FOR GRC OPERATIONS

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Date: 17/01/2026

Executive Summary

This lab focused on developing foundational Linux competencies required for Governance, Risk, and Compliance (GRC) operations. The exercises demonstrated practical skills in file system navigation, permission management, and log analysis. The Key findings indicate that understanding Linux directory structures and permissions is fundamental for accessing configuration files, audit logs, and sensitive system data during compliance assessments.
The results also confirms that improper configuration can introduce material compliance and security risk. All tasks were completed, and evidence has been systematically organized within a secure working directory.

Lab Tasks

Task 1: Linux File System Navigation (Day 1)

Objective Completion & Evidence

Successfully navigated through the Linux file system, executed all required commands, and created the working directory '' **~/grc-lab-week1''**. Screenshots of terminal outputs are referenced in the appendix.

The 5 important system directories and their purposes are:

S/N	System Directory	Purpose	Relevant to GRC
1.	/etc	System configuration files	Misconfiguration here can lead to control failures
2.	/var/log	System and security logs	Primary source of audit and forensic evidence
3.	/home	User home directories	User activity and data exposure risk
4.	/root	Root user home directory	Indicates privileged access
5.	/tmp	Temporary files	High-risk area for misuse or malware

The difference between absolute and relative paths.

Absolute Path: This is a complete path from the root directory to the desired file or directory. It provides the full location of the file in the directory structure, despite the current working directory. For Example: /home/user/documents/audit_evidence.txt

The path starts from the root (/) and specifies the full location to the audit_evidence.txt file.

Relative Path: The relative path, unlike the absolute path, refers to the location of a file or directory in relation to the current working directory. It does not start from the root but instead uses the current location as a reference point. For example: documents/audit_evidencet.txt

The path is relative to the current directory (e.g., if the current directory is /home/user, it points to /home/user/documents/audit_evidence.txt).

Task 2: File Permissions and Security (Day 2)

Objective Completion & Evidence

The audit_evidence directory structure was created, sensitive system file permissions was reviewed, and appropriate permissions on lab files was set. Screenshots with command outputs are appendix in the report for reference.

Understanding permission notation:

1. **Read (r):** This permission allows a user to read the contents of a file or directory. For files users can view the contents of the file. For directories, users can list the files and subdirectories within that directory.

2. **Write (w):** This permission allows a user to modify the contents of a file or directory. For files users can edit the file, which includes adding, deleting, or changing the content. For directories users can create new files or subdirectories, delete existing files, or rename files within that directory.

3. **Execute (x):** This permission allows a user to execute a file or access a directory. For files users can run the file as a program. For directories users can enter the directory and access its contents, provided they also have the read permission.

Octal Notation Explained

Generally, permissions are represented in three sets (owner, group, others). Each right (r, w, x) has a numeric value: r=4, w=2, x=1. The sum defines the permission.

> **644 (-rw-r--r--):** Owner can read/write; group/others can only read. This is appropriate for public policies, non-sensitive reports, etc.
>
> **600 (-rw-------):** Owner can read/write; no access for group/others. This permission is appropriate for confidential evidence.

700 (-rwx------): Owner has full rights (read, write, execute/traverse); no access for others. This permission is appropriate for secure directories containing sensitive evidence.

Task 3: Log Analysis and Extraction (Day 3)

Objective Completion & Evidence

To extract and analyse logs for compliance-relevant security events.

Failed Events Analysis:

The review of log entries containing the keyword "failed" identified a mix of operational and security-relevant events. Most failures were related to service initialization, graphical components, or package validation, which are considered low security risk. From review one failed authentication attempt was observed, which is security-relevant but limited in scope and frequency. This indicates that authentication controls are functioning and that failed access attempts are being properly logged. No evidence of repeated or sustained unauthorized access attempts was identified.

Sudo Activity Analysis:

The analysis of sudo log entries shows that privileged activities are logged. The observed entries primarily reflect configuration checks and administrative or security assessment actions rather than anomalous behaviour. This shows effective privileged access monitoring, a key compliance requirement. There were no indicators of unauthorized or excessive privilege used in the reviewed log sample.

Conclusion: Logging controls are operational and effective, providing visibility into both authentication failures and privileged access. The Observed events do not indicate a material compliance or security risk but should continue to be monitored as part of routine log review procedures.

Count of Sudo Commands:

The grep -I ''sudo'' syslog |head -10 output shows multiple entries related to file validation, permission checks and integrity checks.

The analysis shows that these entries reflect privileged access configuration checks. The privilege activity is been properly logged and this supports compliance with privilege access monitoring requirements.

Count of failed authentication attempts:

The analysis shows 1 failed authentication attempt. The frequency shows that there are no sustained unauthorized access attempts. Our analysis also shows that logging visibility exists, and this is an important aspect of compliance control.

Count of system errors:

The analysis shows ~ 10 system error -related entries, the errors include kernel graphics/VM warnings, GTK theme parser errors, Package signature enforcement errors. These errors are operational.

Identified Security – relevant events:

The analysis revealed the following security relevant events;

a. Privilege access activity (sudo) logged and traceable.
b. Failed authentication attempt detected and recorded.
c. Enforcement of cryptographic standards (SHA-rejection).
d. Authentication subsystem (PAM, LDAP, NIS) status logged.

Our analysis reveals that security-relevant events were logged appropriately. Observed failed authentication activity is low and does not indicate elevated risk. System errors are primarily operational in nature. In all, logging and monitoring controls appears effective and compliant with audit expectations.

TASK 5.3

Executive Security Assessment

Linux Controls, Logging, and Compliance Validation

Prepared for: Chief Information Security Officer (CISO)
Prepared by: GRC Analyst
Scope: Linux system access controls, logging, and audit evidence handling
Environment: Kali Linux

1. **Executive Summary**

This assessment evaluated core Linux security controls that directly support governance, risk management, and compliance objectives, including access control enforcement, privileged activity monitoring, audit logging, and evidence integrity.

The review confirms that:

a. Logging mechanisms are active and effective
b. Privileged (sudo) activity is traceable
c. Authentication failures are detected and recorded
d. File permissions can enforce least privilege

No high-risk security incidents were identified during the review period. Observed issues are primarily operational or environmental and do not indicate compromise. Overall, the control environment demonstrates baseline compliance maturity with opportunities for refinement.

2. Scope

The assessment included:

a. Linux file system navigation and directory validation
b. Review of critical system files (/etc/passwd, /etc/shadow, /etc/sudoers)
c. Permission-based access control testing
d. Log extraction and analysis for security-relevant events
e. Evidence handling and documentation practices

Logging was reviewed using available log files.

3. Key Findings

Privileged Access Monitoring

Observation

Sudo-related activity is consistently logged

Entries primarily reflect configuration validation and security assessment actions

No anomalous or unauthorised privilege escalation observed

Risk Assessment: Low
Privileged access is one of the highest-risk control areas. The ability to trace sudo activity supports insider threat monitoring and audit readiness.

Authentication and Access Events

Observation

a. A small number of failed authentication events were logged
b. No evidence of repeated or sustained failed login attempts
c. Authentication subsystems (PAM, LDAP, NIS) report normal status

Risk Assessment: Low
This demonstrates effective identity monitoring and supports detection of brute-force or misuse scenarios.

System Errors and Operational Events

Observation

a. The System error logs are largely related to:
   Graphics and virtualisation
   Application rendering issues

b. One package installation failure reflected cryptographic policy enforcement (SHA-1 rejection)

Risk Assessment: Low
Operational errors do not currently impact security posture. Cryptographic enforcement errors indicate positive security controls, not weaknesses.

Audit Evidence and Permission Controls

Observation

a. Audit evidence directories were created and structured appropriately
b. Logs are restricted to owner-only access
c. Some evidence directories initially allowed broader read access

Risk Assessment: Low to Moderate
Audit evidence must be protected to ensure integrity and non-repudiation. Overly permissive access increases insider risk.

4. Compliance Implications

The controls observed ensure compliance for:

a. Access control
b. Logging and monitoring
c. Privileged access oversight
d. Security event detection
e. Monitoring and auditability

The environment demonstrates control presence and effectiveness, with no material compliance gaps identified.

6. Recommendations for Improvement

a. **Tighten access to audit evidence directories**

Enforce role-based permissions (e.g., owner-only or restricted groups).

b. **Formalize log review procedures**

Define frequency, ownership, and escalation thresholds.

c. **Document logging architecture**
Explicitly document journald-based logging for audit clarity.

d. **Implement periodic access reviews**
Recertify user and group permissions on sensitive directories.

e. **Standardize evidence handling procedures**

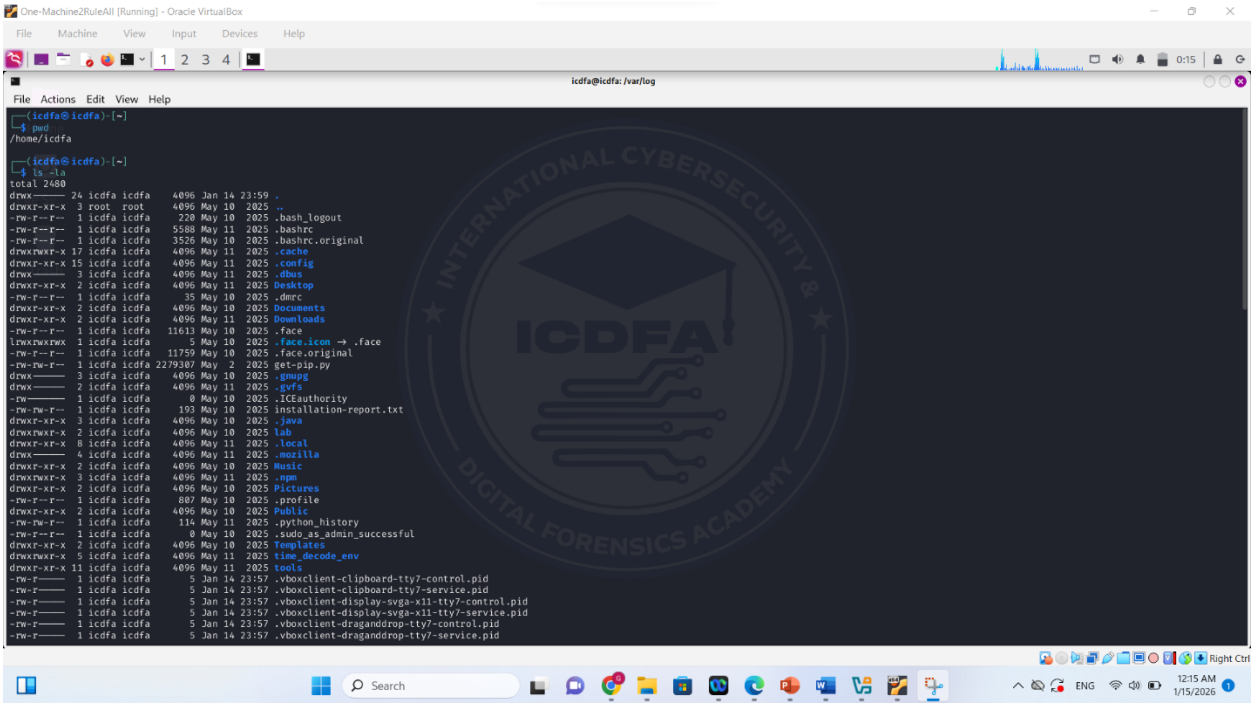
Ensure consistent chain-of-custody practices across systems.

Conclusion

The Linux control environment reviewed demonstrates effective baseline security controls with strong visibility into privileged access and authentication events. Logging and monitoring capabilities meet compliance expectations, and no indicators of compromise were identified. Minor improvements in evidence access control and procedural documentation would further strengthen audit readiness and risk posture.

APPENDIXES FOR SNAPSHOOTS AND AUDIT EVIDENCE

This section covers the snap shoots for commands that were carried out.



Task 1a and 1b

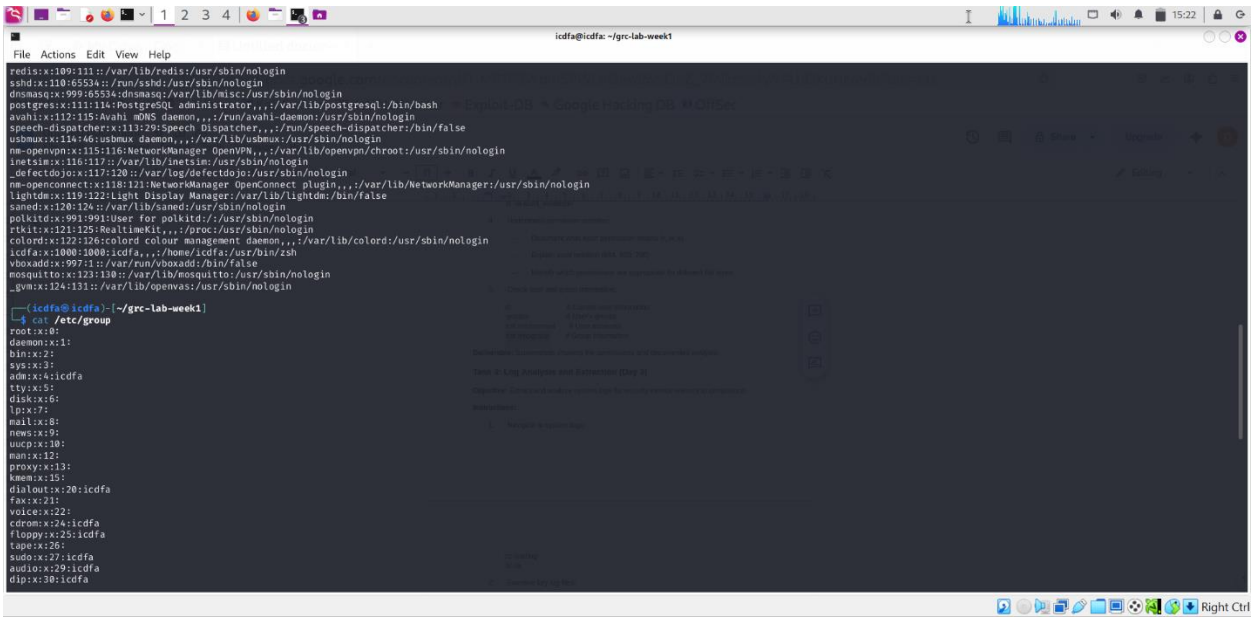
Task 2a

```
┌──(icdfa㉿icdfa)-[~]
└─$ mkdir -p ~/grc-lab-week1

┌──(icdfa㉿icdfa)-[~]
└─$ cd ~/grc-lab-week1

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ pwd
/home/icdfa/grc-lab-week1

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ mkdir -p audit_evidence/{policies,logs,evidence}

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$  ls -la audit_evidence/
total 20
drwxrwxr-x 5 icdfa icdfa 4096 Jan 18 15:07 .
drwxrwxr-x 3 icdfa icdfa 4096 Jan 18 15:07 ..
drwxrwxr-x 2 icdfa icdfa 4096 Jan 18 15:07 evidence
drwxrwxr-x 2 icdfa icdfa 4096 Jan 18 15:07 logs
drwxrwxr-x 2 icdfa icdfa 4096 Jan 18 15:07 policies

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ ls -la /etc/passwd
-rw-r--r-- 1 root root 2777 Oct  4 17:10 /etc/passwd

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$  ls -la /etc/shadow
-rw-r----- 1 root shadow 1272 Oct  4 17:10 /etc/shadow

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ ls -la /etc/sudoers
-r--r----- 1 root root 1714 Feb  9  2025 /etc/sudoers

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ touch audit_evidence/policies/policy_draft.txt

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ chmod 644 audit_evidence/policies/policy_draft.txt

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ touch audit_evidence/evidence/confidential.txt

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ chmod 600 audit_evidence/evidence/confidential.txt
```

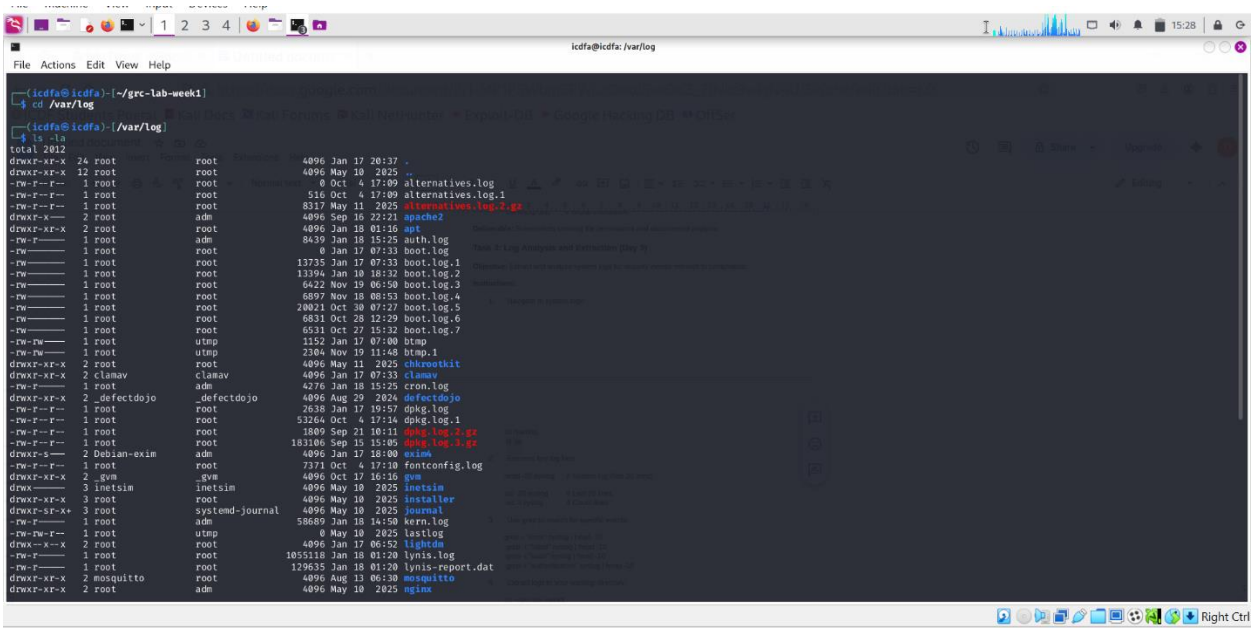
Task 2b

```
┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ mkdir audit_evidence/logs
mkdir: cannot create directory 'audit_evidence/logs': File exists

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ chmod 700 audit_evidence/logs

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ ls -la audit_evidence/
total 20
drwxrwxr-x 5 icdfa icdfa 4096 Jan 18 15:07 .
drwxrwxr-x 3 icdfa icdfa 4096 Jan 18 15:07 ..
drwxrwxr-x 2 icdfa icdfa 4096 Jan 18 15:15 evidence
drwx------ 2 icdfa icdfa 4096 Jan 18 15:07 logs
drwxrwxr-x 2 icdfa icdfa 4096 Jan 18 15:13 policies

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ id
uid=1000(icdfa) gid=1000(icdfa) groups=1000(icdfa),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),101(netdev),113(bluetooth),118(wireshark),123(scanner),128(vboxsf)
,129(kaboxer)

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ groups
icdfa adm dialout cdrom floppy sudo audio dip video plugdev users netdev bluetooth wireshark scanner vboxsf kaboxer

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ cat /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
```

Task 2c



Task 3a



Task 3b

```
icdfa@icdfa: /var/log
File  Actions  Edit  View  Help
└─$ head -20 syslog
2026-01-17T19:57:40.652800+01:00 icdfa systemd[1]: Listening on syslog.socket - Syslog Socket.
2026-01-17T19:57:40.661028+01:00 icdfa systemd[1]: Starting rsyslog.service - System Logging Service...
2026-01-17T19:57:40.642713+01:00 icdfa systemd[1]: Started rsyslog.service - System Logging Service.
2026-01-17T19:57:40.660657+01:00 icdfa rsyslogd: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd.  [v8.2512.0]
2026-01-17T19:57:40.660847+01:00 icdfa rsyslogd: [origin software="rsyslogd" swVersion="8.2512.0" x-pid="138589" x-info="https://www.rsyslog.com"] start
2026-01-17T19:57:40.653241+01:00 icdfa kernel: Linux version 6.12.13-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-14 (Debian 14.2.0-16) 14.2.0, GNU ld (GNU Binutils for Debian) 2.44) #1 SMP PREEMPT_DYNAMIC Kali 6.12.1
3-1kali1 (2025-02-11)
2026-01-17T19:57:40.661380+01:00 icdfa kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.12.13-amd64 root=UUID=a5089483-4c45-4934-865d-711741236ab9 ro quiet splash
2026-01-17T19:57:40.661386+01:00 icdfa kernel: BIOS-provided physical RAM map:
2026-01-17T19:57:40.661387+01:00 icdfa kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] usable
2026-01-17T19:57:40.661389+01:00 icdfa kernel: BIOS-e820: [mem 0x000000000009fc00-0x000000000009ffff] reserved
2026-01-17T19:57:40.661391+01:00 icdfa kernel: BIOS-e820: [mem 0x00000000000f0000-0x00000000000fffff] reserved
2026-01-17T19:57:40.661392+01:00 icdfa kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000dffeffff] usable
2026-01-17T19:57:40.661396+01:00 icdfa kernel: BIOS-e820: [mem 0x00000000dff00000-0x00000000dfffffff] ACPI data
2026-01-17T19:57:40.661398+01:00 icdfa kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved
2026-01-17T19:57:40.661400+01:00 icdfa kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved
2026-01-17T19:57:40.661401+01:00 icdfa kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved
2026-01-17T19:57:40.661403+01:00 icdfa kernel: BIOS-e820: [mem 0x0000000100000000-0x00000001b8cfffff] usable
2026-01-17T19:57:40.661404+01:00 icdfa kernel: NX (Execute Disable) protection: active
2026-01-17T19:57:40.661406+01:00 icdfa kernel: APIC: Static calls initialized
2026-01-17T19:57:40.661410+01:00 icdfa kernel: SMBIOS 2.5 present.

┌──(icdfa㉿icdfa)-[/var/log]
└─$ tail -20 syslog
2026-01-18T14:50:42.398629+01:00 icdfa kernel: 13:50:41.362251 timesync vgsvcTimeSyncWorker: Radical host time change: 43 889 819 000 000ns (HostNow=1 768 744 241 362 000 000
2026-01-18T14:50:42.399218+01:00 icdfa kernel: e1000: eth1 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
2026-01-18T14:50:42.402968+01:00 icdfa NetworkManager[592]: <info>  [1768744242.3995] device (eth1): carrier: link connected
2026-01-18T14:54:33.244502+01:00 icdfa nautilus[158800]: Broken accounting of active state for widget 0x5610db78c6f0(GtkPopoverMenu)
2026-01-18T14:55:01.200248+01:00 icdfa CRON[280281]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2026-01-18T14:55:03.040970+01:00 icdfa dbus-daemon[526]: [system] Activating via systemd: service name='org.freedesktop.hostname1' unit='dbus-org.freedesktop.hostname1.service' requested by ':1.83' (uid=1000 pid=15880
0 comm="/usr/bin/nautilus --gapplication-service")
2026-01-18T14:55:03.063680+01:00 icdfa systemd[1]: Starting systemd-hostnamed.service - Hostname Service...
2026-01-18T14:55:03.421666+01:00 icdfa systemd[1]: Started systemd-hostnamed.service - Hostname Service.
2026-01-18T14:55:03.440759+01:00 icdfa dbus-daemon[526]: [system] Successfully activated service 'org.freedesktop.hostname1'
2026-01-18T14:55:24.277955+01:00 icdfa xdg-desktop-por[2411]: A backend call failed: Inhibiting other than idle not supported
2026-01-18T14:55:34.547500+01:00 icdfa systemd[1]: systemd-hostnamed.service: Deactivated successfully.
2026-01-18T15:00:01.266701+01:00 icdfa CRON[282829]: (root) CMD (   test -x /usr/sbin/tigercron && { [ -r "$DEFAULT" ] && . "$DEFAULT" ; nice -n$NICETIGER /usr/sbin/tigercron -q ; })
2026-01-18T15:05:01.588936+01:00 icdfa CRON[285419]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2026-01-18T15:09:01.619857+01:00 icdfa systemd[1]: Starting phpsessionclean.service - Clean php session files...
2026-01-18T15:09:01.627604+01:00 icdfa CRON[287437]: (root) CMD (   [ -x /usr/lib/php/sessionclean ] && if [ ! -d /run/systemd/system ]; then /usr/lib/php/sessionclean; fi)
2026-01-18T15:09:01.864768+01:00 icdfa systemd[1]: phpsessionclean.service: Deactivated successfully.
2026-01-18T15:09:01.866031+01:00 icdfa systemd[1]: Finished phpsessionclean.service - Clean php session files.
2026-01-18T15:15:01.689776+01:00 icdfa CRON[290434]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
2026-01-18T15:17:01.736224+01:00 icdfa CRON[291422]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
2026-01-18T15:25:01.786551+01:00 icdfa CRON[295392]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
```

Task 3c

```
2026-01-18T15:17:01.736224+01:00 icdfa CRON[291422]: (root) CMD (cd / && run-parts --r
2026-01-18T15:25:01.786551+01:00 icdfa CRON[295392]: (root) CMD (command -v debian-sa1

┌──(icdfa㉿icdfa)-[/var/log]
└─$ wc -l syslog
1454 syslog

┌──(icdfa㉿icdfa)-[/var/log]
```

Task 3d

```
icdfa@icdfa: /var/log
File  Actions  Edit  View  Help
└─$ grep -i "error" syslog | head -10
2026-01-17T19:57:40.662850+01:00 icdfa kernel: RAS: Correctable Errors collector initialized.
2026-01-17T19:57:40.663197+01:00 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
2026-01-17T19:57:40.663199+01:00 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.
2026-01-17T19:57:40.663201+01:00 icdfa kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.
2026-01-17T20:37:27.357442+01:00 icdfa nautilus[158800]: Theme parser error: gtk.css:3995:3-22: No property named "-gtk-outline-radius"
2026-01-17T20:37:27.361595+01:00 icdfa nautilus[158800]: Theme parser error: gtk.css:4112:3-29: No property named "-GtkWidget-window-dragging"
2026-01-17T20:37:27.362173+01:00 icdfa nautilus[158800]: Theme parser error: gtk.css:4126:9-37: No property named "-gtk-outline-top-left-radius"
2026-01-17T20:37:27.362657+01:00 icdfa nautilus[158800]: Theme parser error: gtk.css:4128:9-38: No property named "-gtk-outline-top-right-radius"
2026-01-17T20:37:27.362930+01:00 icdfa nautilus[158800]: Theme parser error: gtk.css:4131:9-40: No property named "-gtk-outline-bottom-left-radius"
2026-01-17T20:37:27.363402+01:00 icdfa nautilus[158800]: Theme parser error: gtk.css:4133:9-41: No property named "-gtk-outline-bottom-right-radius"

┌──(icdfa㉿icdfa)-[/var/log]
└─$ grep -i "failed" syslog | head -10
2026-01-17T19:57:40.663480+01:00 icdfa kernel: 05:52:06.285825 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
2026-01-17T20:29:39.232779+01:00 icdfa dbus-daemon[526]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.service': Unit dbus-org.freedesktop.Avahi.service not found.
2026-01-17T20:37:33.382760+01:00 icdfa xdg-desktop-por[2411]: A backend call failed: Inhibiting other than idle not supported
2026-01-17T20:37:36.032573+01:00 icdfa org.gnome.NautilusPreviewer[158841]: libEGL warning: egl: failed to create dri2 screen
2026-01-17T20:37:36.033083+01:00 icdfa org.gnome.NautilusPreviewer[158841]: libEGL warning: DRI2: failed to create screen
2026-01-17T20:37:36.242089+01:00 icdfa org.gnome.NautilusPreviewer[158841]: MESA: error: ZINK: failed to choose pdev
2026-01-17T20:37:36.248695+01:00 icdfa org.gnome.NautilusPreviewer[158841]: libEGL warning: egl: failed to create dri2 screen
2026-01-17T20:37:36.249016+01:00 icdfa org.gnome.NautilusPreviewer[158841]: libEGL warning: DRI2: failed to create screen
2026-01-18T01:14:16.374763+01:00 icdfa lynis[162126]: - Logging failed login attempts [ DISABLED ]
2026-01-18T01:15:56.179566+01:00 icdfa lynis[214643]: W: Failed to fetch https://artifacts.elastic.co/packages/8.x/apt/dists/stable/InRelease  Sub-process /usr/bin/sqv returned an error code (1), error message is: Sig
ning key on 46895ACC8548582C1A2699A9D27D666CD88E42B4 is not bound:              No binding signature at time 2026-01-13T11:53:41Z    because: Policy rejected non-revocation signature (PositiveCertification) requiring sec
ond pre-image resistance    because: SHA1 is not considered secure since 2026-01-01T00:00:00Z

┌──(icdfa㉿icdfa)-[/var/log]
└─$ grep -i "sudo" syslog | head -10
2026-01-18T01:14:11.993837+01:00 icdfa lynis[162126]: - Sudoers file(s) [ FOUND ]
2026-01-18T01:14:12.131777+01:00 icdfa lynis[162126]: - Permissions for directory: /etc/sudoers.d [ WARNING ]
2026-01-18T01:14:12.191396+01:00 icdfa lynis[162126]: - Permissions for: /etc/sudoers [ OK ]
2026-01-18T01:14:12.300840+01:00 icdfa lynis[162126]: - Permissions for: /etc/sudoers.d/kali-grant-root [ OK ]
2026-01-18T01:14:12.374237+01:00 icdfa lynis[162126]: - Permissions for: /etc/sudoers.d/ospd-openvas [ OK ]
2026-01-18T01:14:12.455939+01:00 icdfa lynis[162126]: - Permissions for: /etc/sudoers.d/README [ OK ]

┌──(icdfa㉿icdfa)-[/var/log]
└─$ grep -i "authentication" syslog | head -10
2026-01-18T01:13:39.066203+01:00 icdfa lynis[162126]: - Authentication:
2026-01-18T01:13:39.131620+01:00 icdfa lynis[162126]: - PAM (Pluggable Authentication Modules):
2026-01-18T01:14:10.432286+01:00 icdfa lynis[162126]: [+] Users, Groups and Authentication
2026-01-18T01:14:11.770812+01:00 icdfa lynis[162126]: - NIS+ authentication support [ NOT ENABLED ]
2026-01-18T01:14:11.900099+01:00 icdfa lynis[162126]: - NIS authentication support [ NOT ENABLED ]
2026-01-18T01:14:15.199528+01:00 icdfa lynis[162126]: - Checking Linux single user mode authentication [ OK ]
2026-01-18T01:14:15.845488+01:00 icdfa lynis[162126]: - LDAP authentication support [ NOT ENABLED ]
2026-01-18T01:20:30.942927+01:00 icdfa lynis[162126]:   * Install fail2ban to automatically ban hosts that commit multiple authentication errors. [DEB-0880]
```

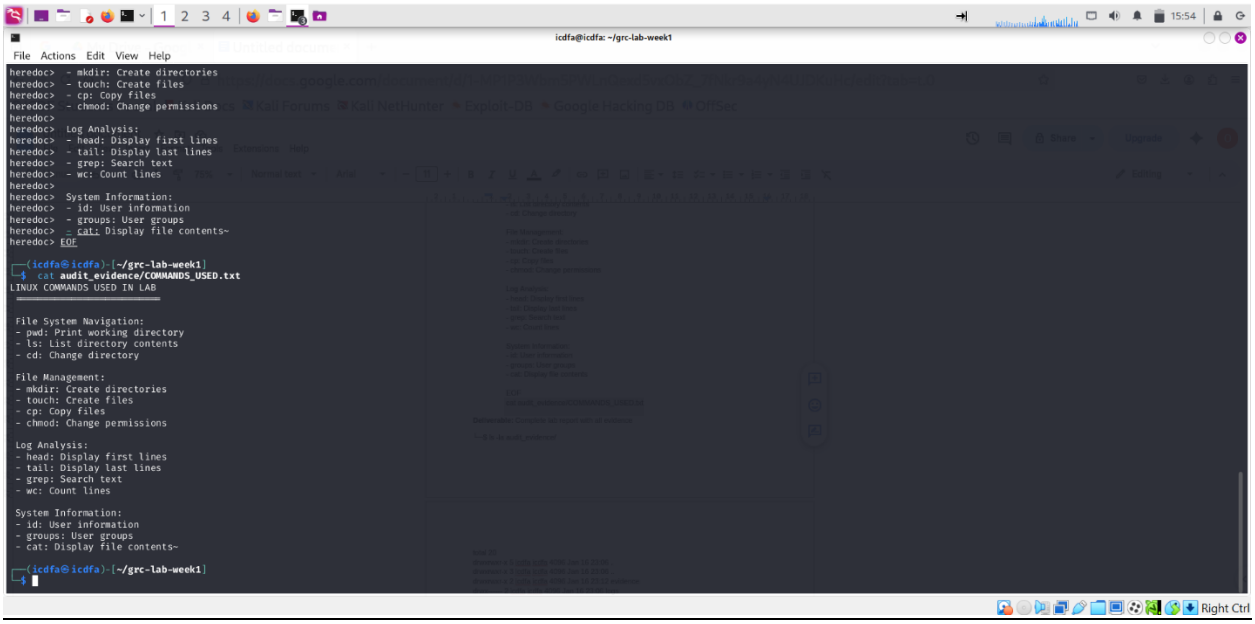
Task 5a

```
icdfa@icdfa: ~/grc-lab-week1
File  Actions  Edit  View  Help
2026-01-18T01:20:30.942927+01:00 icdfa lynis[162126]:   * Install fail2ban to automatically ban hosts that commit multiple authentication errors. [DEB-0880]
2026-01-18T01:20:31.649973+01:00 icdfa lynis[162126]:     * Article: Linux password security: hashing rounds: https://linux-audit.com/authentication/configure-the-minimum-password-length-on-linux-systems/

┌──(icdfa㉿icdfa)-[/var/log]
└─$ cd ~/grc-lab-week1

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ cp /var/log/syslog audit_evidence/logs/syslog_sample.txt

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ cp /var/log/auth.log audit_evidence/logs/auth_sample.txt 2>/dev/null || echo "auth.log not available"

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ grep "sudo" audit_evidence/logs/syslog_sample.txt | wc -l
5

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ grep -i "failed" audit_evidence/logs/syslog_sample.txt | head -5
2026-01-17T19:57:40.663480+01:00 icdfa kernel: 05:52:06.285825 main      rtldrNativeLoad: dlopen('libwayland-client.so', RTLD_NOW | RTLD_LOCAL) failed: libwayland-client.so:
2026-01-17T20:29:39.232779+01:00 icdfa dbus-daemon[526]: [system] Activation via systemd failed for unit 'dbus-org.freedesktop.Avahi.service': Unit dbus-org.freedesktop.Avahi.service not found.
2026-01-17T20:37:33.382760+01:00 icdfa xdg-desktop-por[2411]: A backend call failed: Inhibiting other than idle not supported
2026-01-17T20:37:36.032573+01:00 icdfa org.gnome.NautilusPreviewer[158841]: libEGL warning: egl: failed to create dri2 screen
2026-01-17T20:37:36.033083+01:00 icdfa org.gnome.NautilusPreviewer[158841]: libEGL warning: DRI2: failed to create screen

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ ls -la audit_evidence/
total 20
drwxrwxr-x 5 icdfa icdfa 4096 Jan 18 15:07 .
drwxrwxr-x 3 icdfa icdfa 4096 Jan 18 15:07 ..
drwxrwxr-x 2 icdfa icdfa 4096 Jan 18 15:15 evidence
drwx------ 2 icdfa icdfa 4096 Jan 18 15:43 logs
drwxrwxr-x 2 icdfa icdfa 4096 Jan 18 15:13 policies

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ find audit_evidence/ -type f
audit_evidence/evidence/confidential.txt
audit_evidence/policies/policy_draft.txt
audit_evidence/logs/auth_sample.txt
audit_evidence/logs/syslog_sample.txt

┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ cat > audit_evidence/COMMANDS_USED.txt << 'EOF'
 LINUX COMMANDS USED IN LAB
 ═══════════════════════

 File System Navigation:
```

Task 5b

```
icdfa@icdfa: ~/grc-lab-week1
File  Actions  Edit  View  Help
┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ cat > audit_evidence/COMMANDS_USED.txt << 'EOF'
 LINUX COMMANDS USED IN LAB
 ═══════════════════════

 File System Navigation:
 - pwd: Print working directory
 - ls: List directory contents
 - cd: Change directory

 File Management:
 - mkdir: Create directories
 - touch: Create files
 - cp: Copy files
 - chmod: Change permissions

 Log Analysis:
 - head: Display first lines
 - tail: Display last lines
 - grep: Search text
 - wc: Count lines

 System Information:
 - id: User information
 - groups: User groups
 - cat: Display file contents

 EOF
heredoc> LINUX COMMANDS USED IN LAB
 ═══════════════════════

 File System Navigation:
 - pwd: Print working directory
 - ls: List directory contents
 - cd: Change directory

 File Management:
 - mkdir: Create directories
 - touch: Create files
 - cp: Copy files
 - chmod: Change permissions

 Log Analysis:
 - head: Display first lines
 - tail: Display last lines
```

Task 5c


icdfa@icdfa: ~/grc-lab-week1
File Actions Edit View Help
heredoc> - mkdir: Create directories
heredoc> - touch: Create files
heredoc> - cp: Copy files
heredoc> - chmod: Change permissions
heredoc>
heredoc> Log Analysis:
heredoc> - head: Display first lines
heredoc> - tail: Display last lines
heredoc> - grep: Search text
heredoc> - wc: Count lines
heredoc>
heredoc> System Information:
heredoc> - id: User information
heredoc> - groups: User groups
heredoc> - cat: Display file contents~
heredoc> EOF
┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$ cat audit_evidence/COMMANDS_USED.txt
LINUX COMMANDS USED IN LAB
File System Navigation:
- pwd: Print working directory
- ls: List directory contents
- cd: Change directory
File Management:
- mkdir: Create directories
- touch: Create files
- cp: Copy files
- chmod: Change permissions
Log Analysis:
- head: Display first lines
- tail: Display last lines
- grep: Search text
- wc: Count lines
System Information:
- id: User information
- groups: User groups
- cat: Display file contents~
┌──(icdfa㉿icdfa)-[~/grc-lab-week1]
└─$
Right Ctrl